



Group 7

Hierarchical Clustering

Building a Family Tree for Your Data

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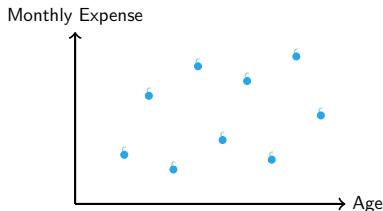
Outline

- 1 Introduction - Clustering Basics
- 2 Transition to Hierarchical Clustering
- 3 Hierarchical Clustering - Introduction
- 4 Types of Hierarchical Clustering
 - Agglomerative Clustering
 - Divisive Clustering
- 5 Linkage Criteria

A Problem to Think About

Example (Superstore Customer Problem)

A large superstore wants to understand their 10,000 customers better. They have data about each customer:

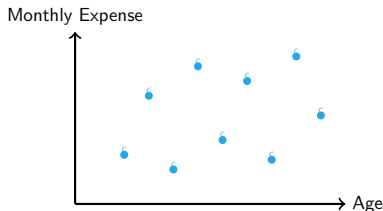


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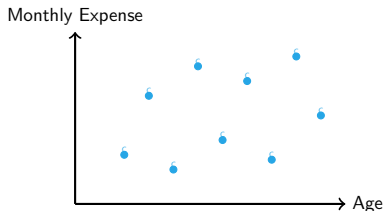


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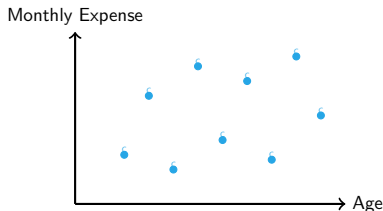


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The Question

How can they group these customers to create targeted offers for each group?

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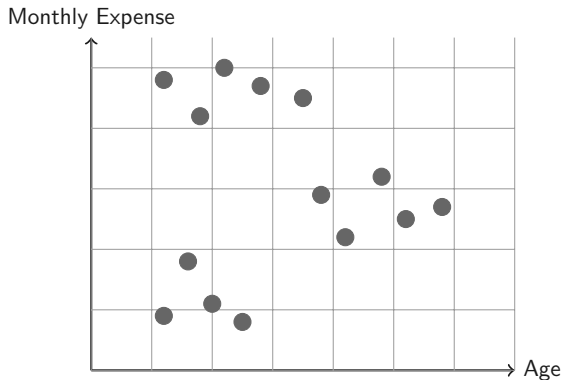
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Think

Manually grouping 10,000 customers? **Impossible!**

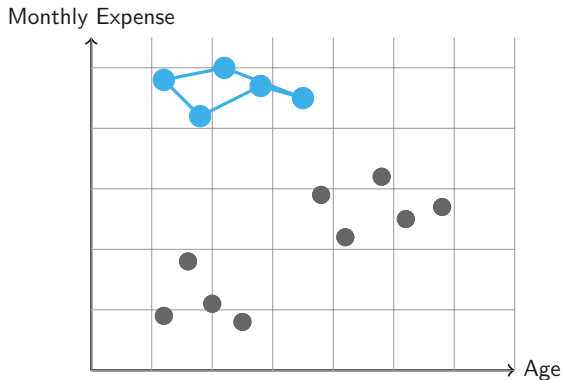
Solution: Clustering!

- **Clustering** automatically groups similar customers



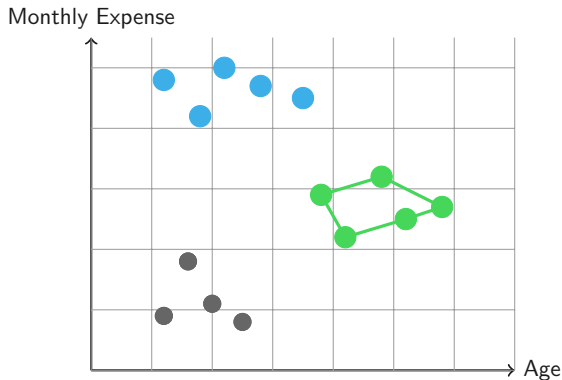
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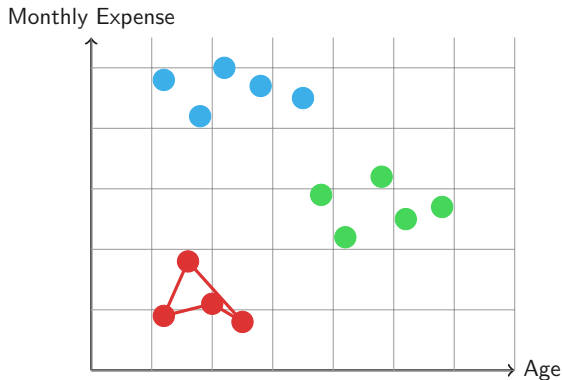
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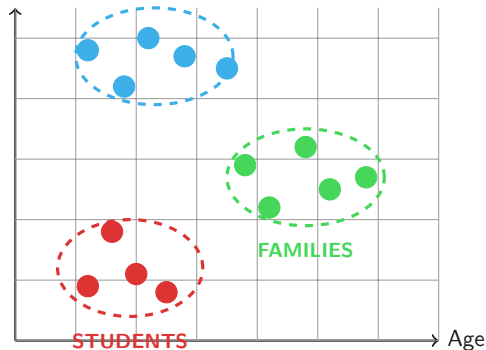
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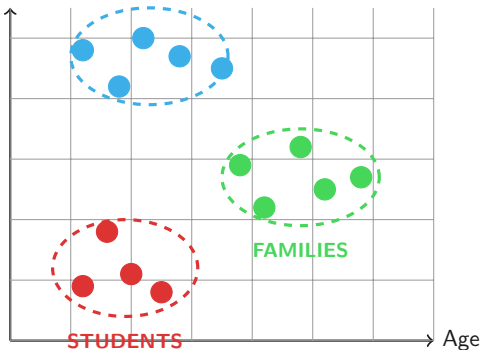
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Solution: Clustering!

- **Clustering** automatically groups similar customers
- Each group = **Cluster**
- Boundaries define each cluster with **meaningful names**

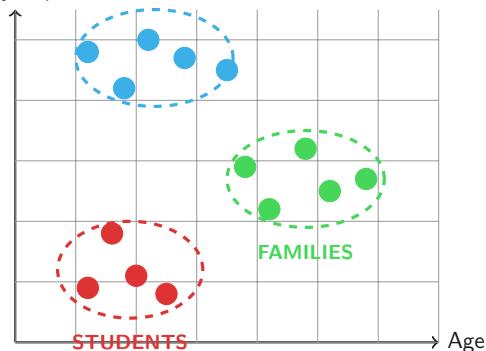
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Solution: Clustering!

- **Clustering** automatically groups similar customers
- Each group = **Cluster**
- Boundaries define each cluster with **meaningful names**
- Customers in same cluster have similar:
 - Age
 - Spending habits
 - Preferences

Monthly Expense **PREMIUM**

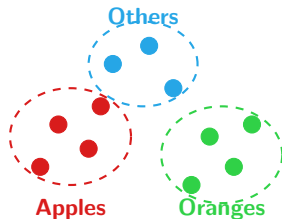


Each cluster contains customers with similar characteristics!

What is Clustering?

Definition

Clustering is the process of **dividing the datasets** into groups, consisting of **similar data-points**.



Analogy

Like sorting a fruit basket: all **apples** go in one bowl, all **oranges** in another.

But...

But...

Regular clustering gives us **groups**...

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But it doesn't tell us **how groups are related** to each other!

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Regular clustering gives us **groups**...

But it doesn't tell us **how groups are related** to each other!

We need something more...

Here Comes...



Hierarchical Clustering!

DONE!

What we covered?

Real-life Problem Example

Clustering Fundamentals

The Need of Hierarchical Clustering

Over to **Speaker 2**

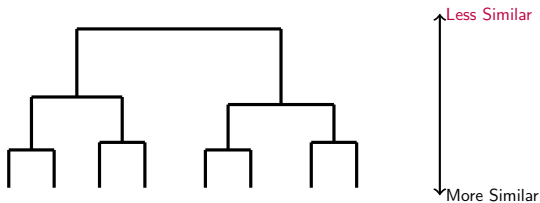
Definition of Hierarchical Clustering

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Hierarchical Clustering

Hierarchical cluster analysis or **HCA** is a method of cluster analysis that builds a **tree structure** (*dendrogram*) from data similarities.

Dendrogram



Types of Hierarchical Clustering

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Agglomerative

Bottom-Up Approach

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Agglomerative

Bottom-Up Approach

Divisive

Top-Down Approach

Agglomerative Clustering (Bottom-Up)

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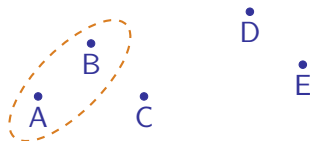
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Agglomerative Clustering (Bottom-Up)

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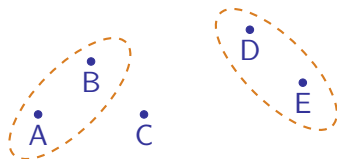
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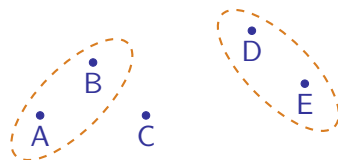
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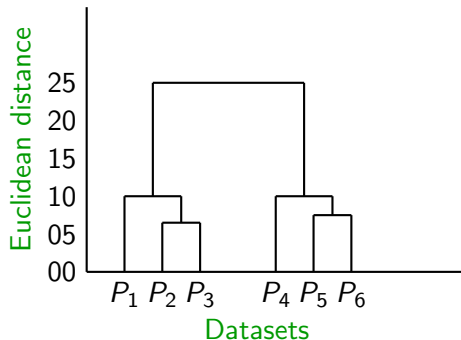
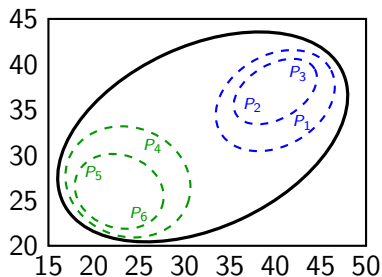
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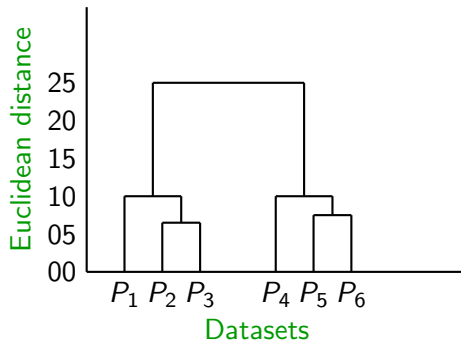
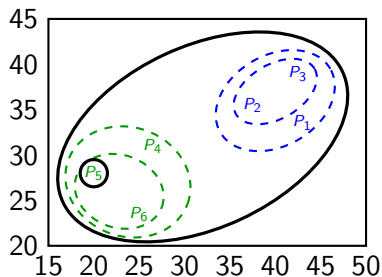


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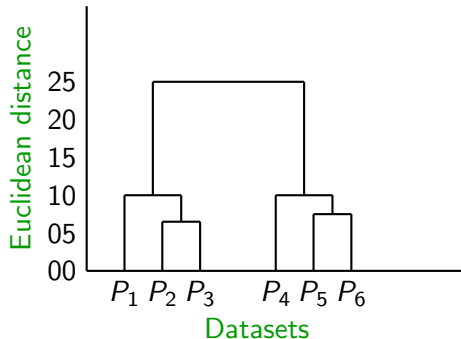
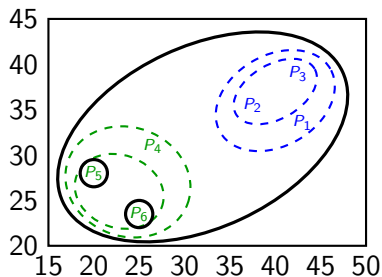
Agglomerative Clustering Example



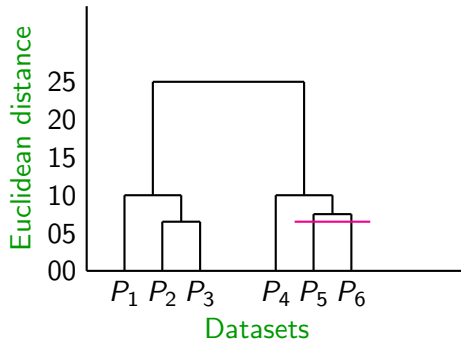
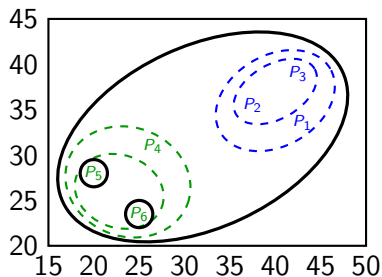
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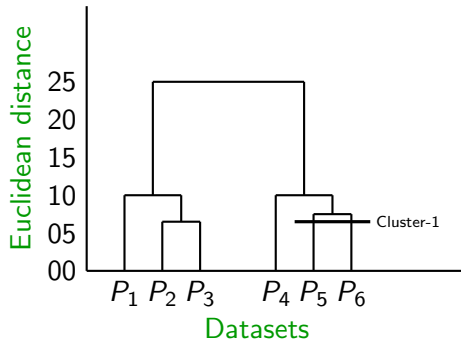
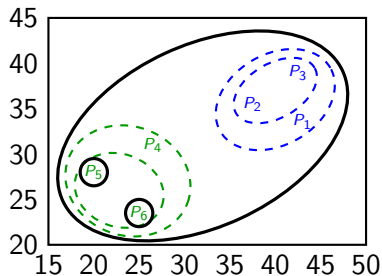
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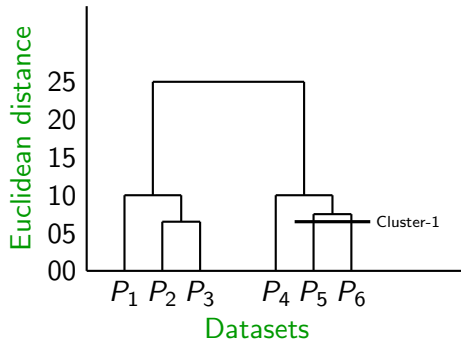
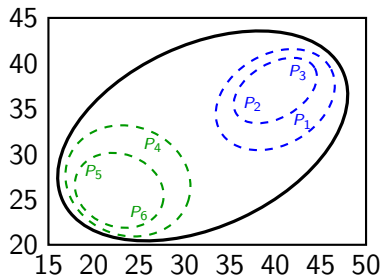
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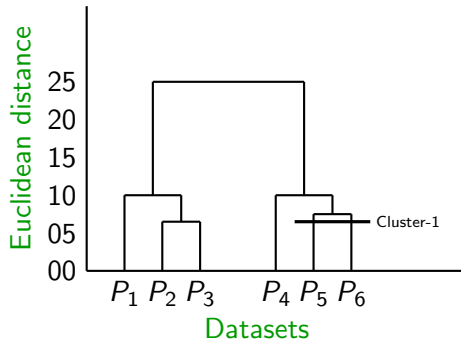
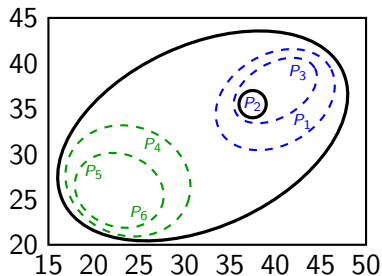
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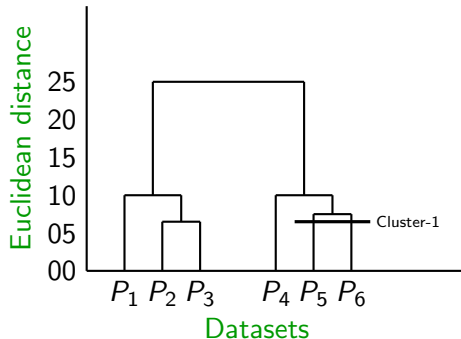
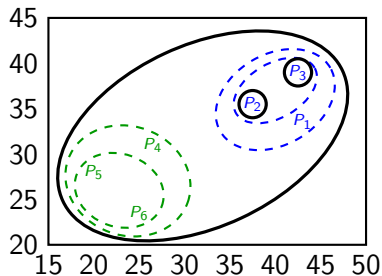
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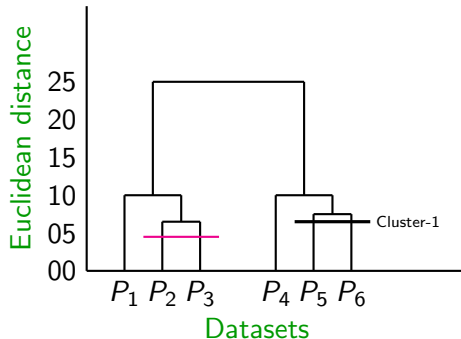
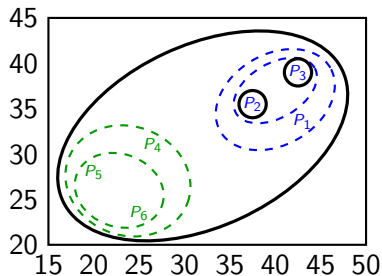
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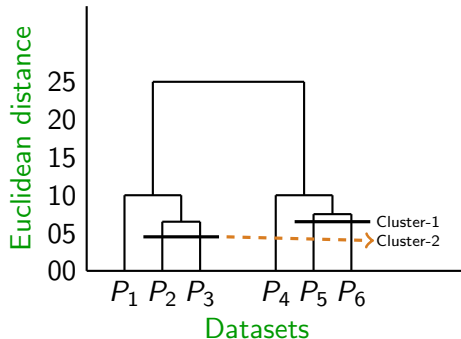
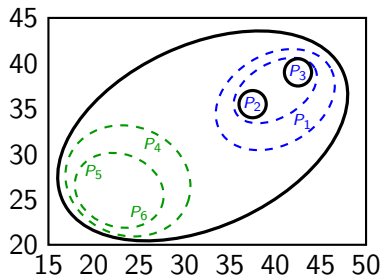
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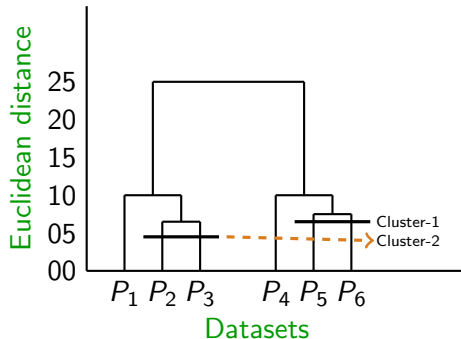
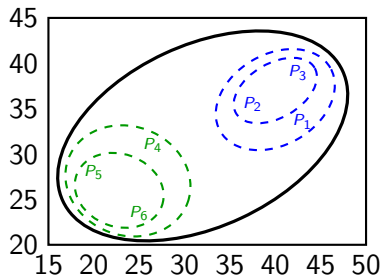
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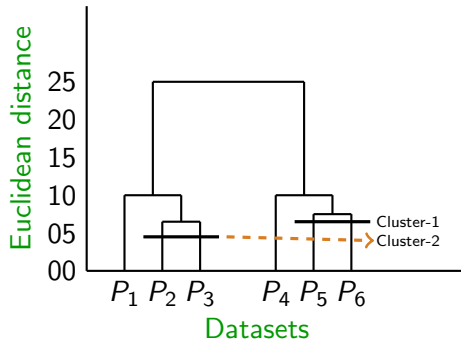
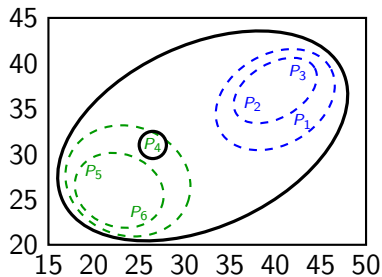
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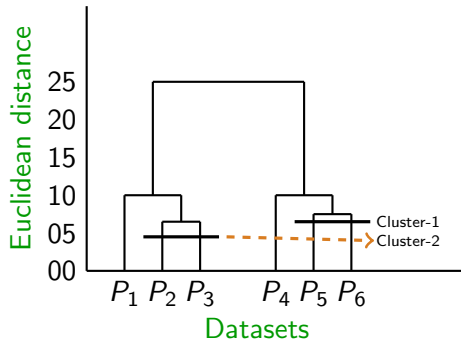
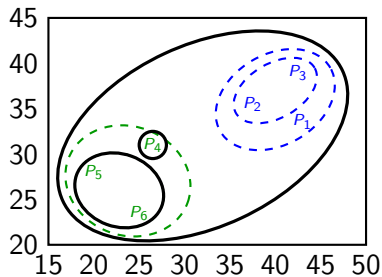
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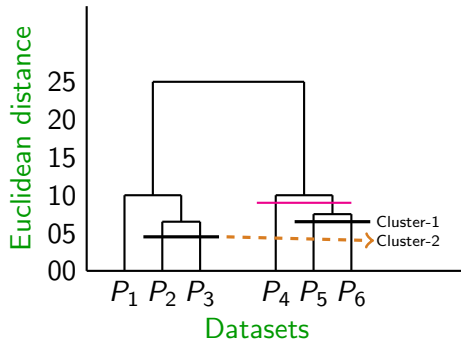
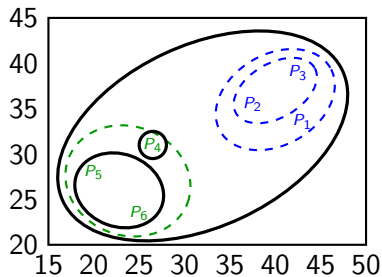
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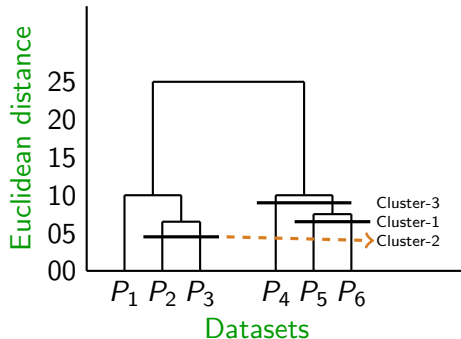
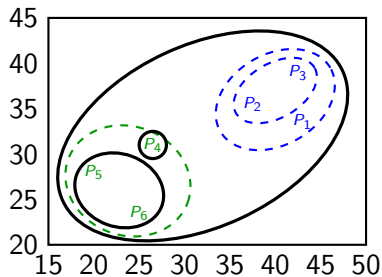
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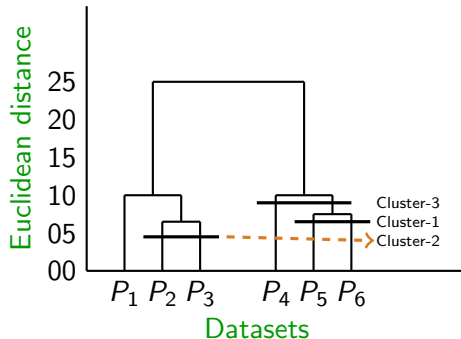
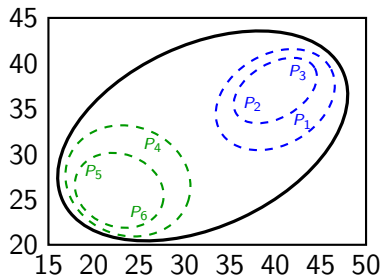
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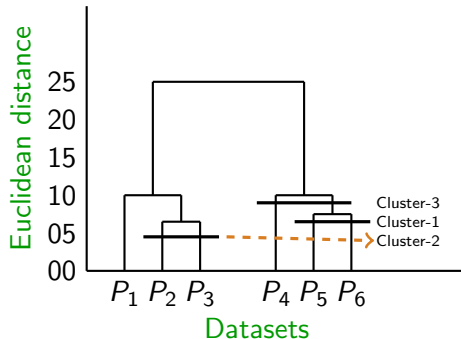
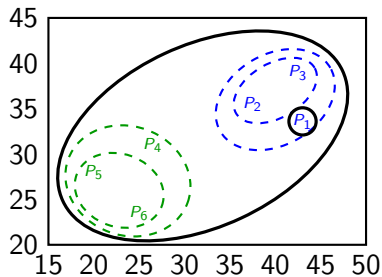
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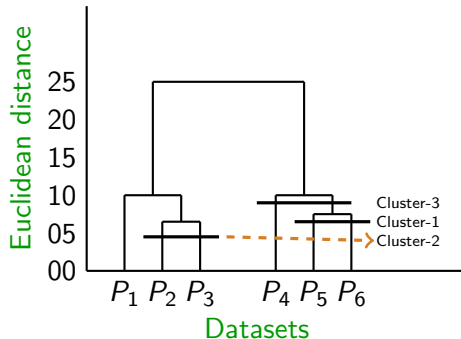
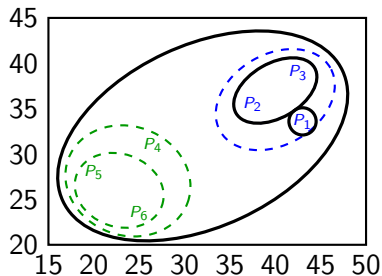
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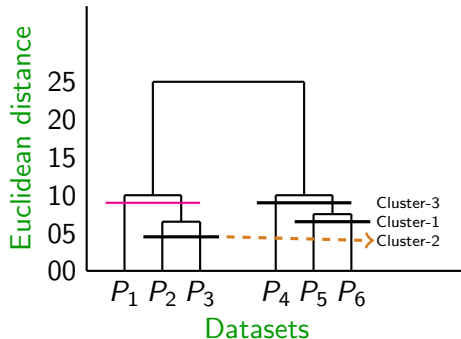
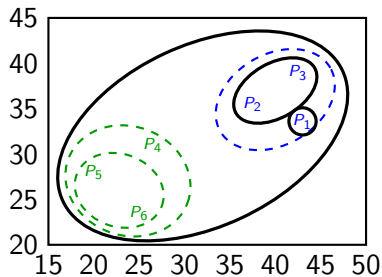
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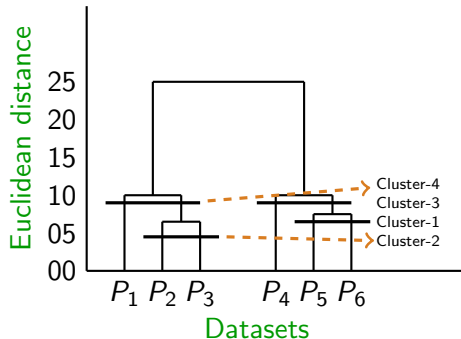
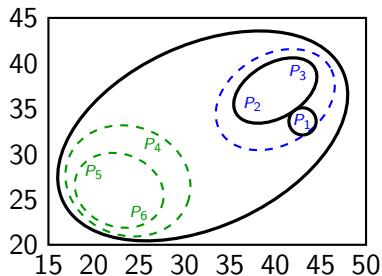
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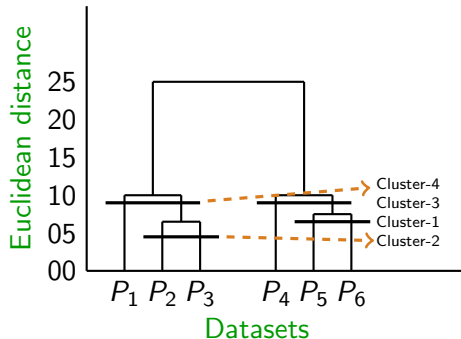
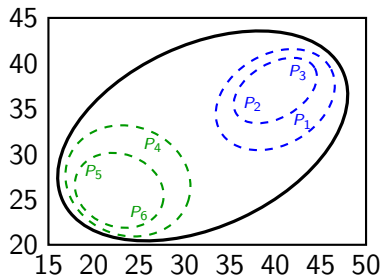
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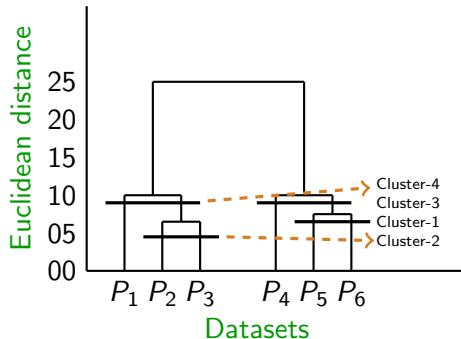
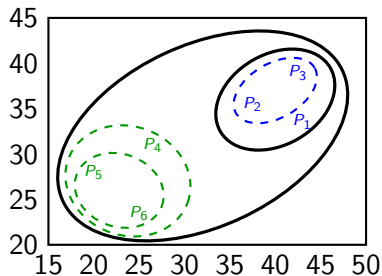
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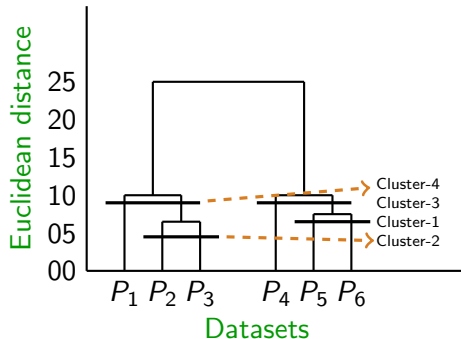
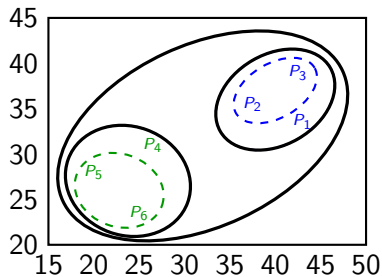
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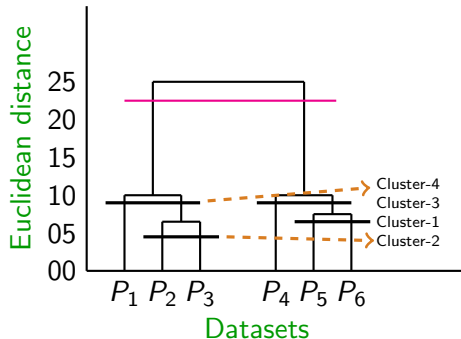
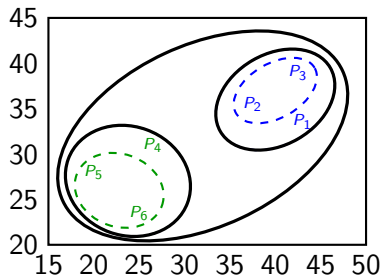
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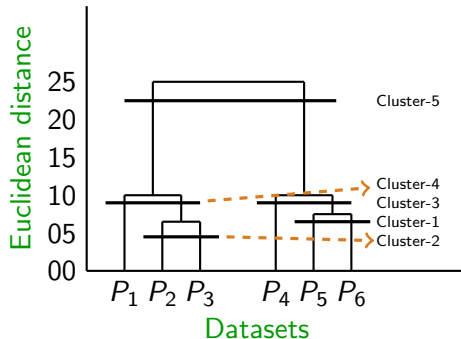
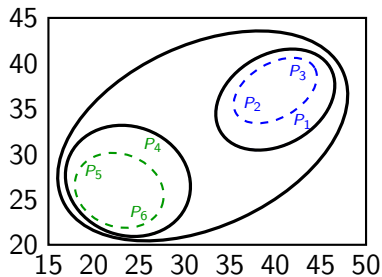
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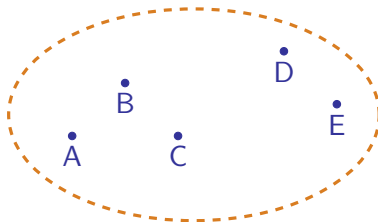
Divisive Clustering

Divisive (Top-Down)

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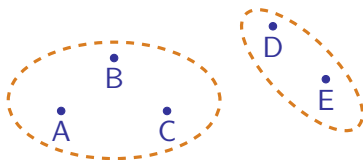
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Divisive Clustering

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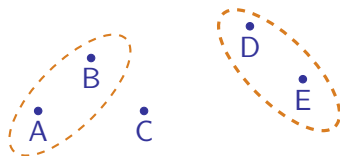
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- 2 **Split** a cluster into two smaller ones.



Divisive Clustering

Divisive (Top-Down)

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- 2 **Split** a cluster into two smaller ones.
- 3 Repeat until each point is its own cluster.



and so on...

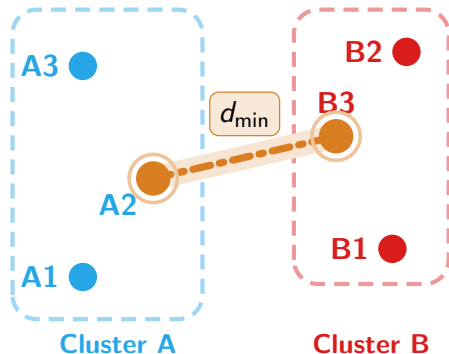
Which is more common?

Agglomerative is more popular because it's computationally simpler.

How Do We Define “Closeness”? (Linkage Methods)

- **Single Linkage (Min):**

- Closest pair of points.
- Can lead to “chaining” (long, thin clusters).



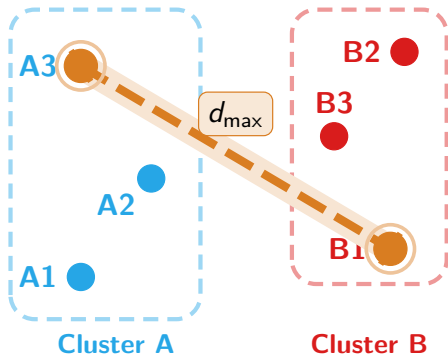
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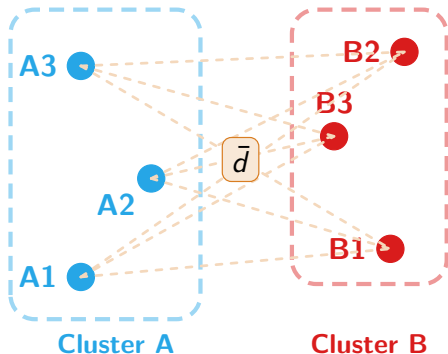
- **Complete Linkage (Max):**

- Furthest pair of points.
- *Produces compact, spherical clusters.*



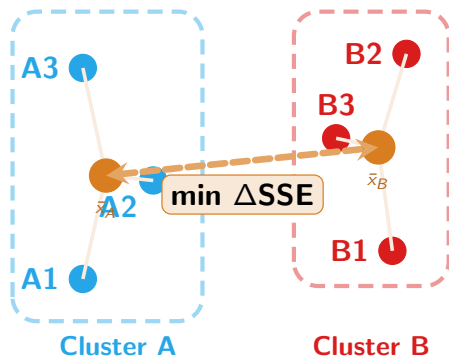
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- **Complete Linkage (Max):**
 - Furthest pair of points.
 - Produces compact, spherical clusters.
- **Average Linkage (Mean):**
 - Average of all pairs.
 - A robust “middle ground” approach.



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- **Single Linkage (Min):**
 - Closest pair of points.
 - *Can lead to “chaining” (long, thin clusters).*
- **Complete Linkage (Max):**
 - Furthest pair of points.
 - *Produces compact, spherical clusters.*
- **Average Linkage (Mean):**
 - Average of all pairs.
 - *A robust “middle ground” approach.*
- **Ward’s Method:**
 - Minimizes within-cluster variance.
 - *Most popular for balanced data.*



Single Linkage Clustering

Single linkage distance:

$$D(C_i, C_j) = \min_{x \in C_i, y \in C_j} d(x, y)$$

Distance Matrix

	A	B	C	D
A	0	2	6	10
B	2	0	5	9
C	6	5	0	4
D	10	9	4	0

Dendrogram

A B C D

Single Linkage Clustering

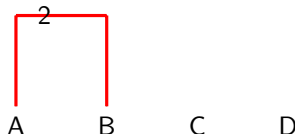
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Dendrogram



Single Linkage Clustering

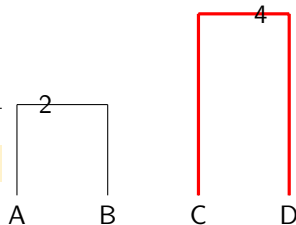
Single linkage distance:

$$D(C_i, C_j) = \min_{x \in C_i, y \in C_j} d(x, y)$$

Distance Matrix

	<i>AB</i>	<i>C</i>	<i>D</i>
<i>AB</i>	0	$\min(6, 5) = 5$	$\min(10, 9) = 9$
<i>C</i>	5	0	4
<i>D</i>	9	4	0

Dendrogram



Single Linkage Clustering

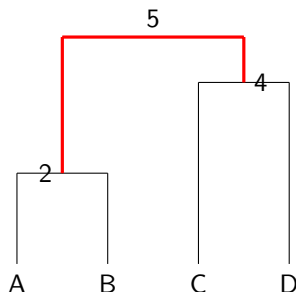
Single linkage distance:

$$D(C_i, C_j) = \min_{x \in C_i, y \in C_j} d(x, y)$$

Distance Matrix

	<i>AB</i>	<i>CD</i>
<i>AB</i>	0	$\min(5, 9) = 5$
<i>CD</i>	5	0

Dendrogram



Complete Linkage Clustering

Complete linkage distance:

$$D(C_i, C_j) = \max_{x \in C_i, y \in C_j} d(x, y)$$

Distance Matrix

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
<i>A</i>	0	2	6	10
<i>B</i>	2	0	5	9
<i>C</i>	6	5	0	4
<i>D</i>	10	9	4	0

Dendrogram

A B C D

Complete Linkage Clustering

Complete linkage distance:

$$D(C_i, C_j) = \max_{x \in C_i, y \in C_j} d(x, y)$$

Distance Matrix

	A	B	C	D
A	0	2	6	10
B	2	0	5	9
C	6	5	0	4
D	10	9	4	0

Dendrogram



Complete Linkage Clustering

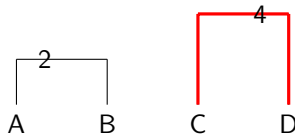
Complete linkage distance:

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Distance Matrix

	<i>AB</i>	<i>C</i>	<i>D</i>
<i>AB</i>	0	$\max(6, 5) = 6$	$\max(10, 9) = 10$
<i>C</i>	6	0	4
<i>D</i>	10	4	0

Dendrogram



Complete Linkage Clustering

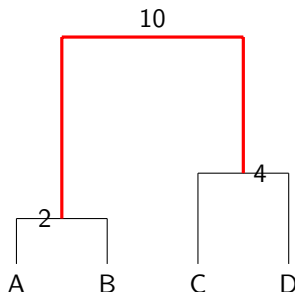
Complete linkage distance:

$$D(C_i, C_j) = \max_{x \in C_i, y \in C_j} d(x, y)$$

Distance Matrix

	<i>AB</i>	<i>CD</i>
<i>AB</i>	0	$\max(6, 10) = 10$
<i>CD</i>	10	0

Dendrogram



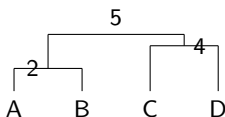
Comparison: Single vs Complete Linkage

Original Distance Matrix

	A	B	C	D
A	0	2	6	10
B	2	0	5	9
C	6	5	0	4
D	10	9	4	0

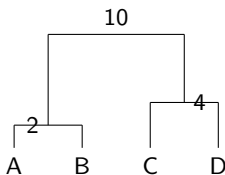
Single Linkage

$$D(C_i, C_j) = \min_{x \in C_i, y \in C_j} d(x, y)$$



Complete Linkage

$$D(C_i, C_j) = \max_{x \in C_i, y \in C_j} d(x, y)$$



Same initial merges (A-B at 2, C-D at 4), but final merge height: 5 (single) vs. 10 (complete).

Average Linkage Clustering

Average linkage distance:

$$D(C_i, C_j) = \frac{1}{|C_i| \cdot |C_j|} \sum_{x \in C_i} \sum_{y \in C_j} d(x, y)$$

Distance Matrix

	A	B	C	D
A	0	2	6	10
B	2	0	5	9
C	6	5	0	4
D	10	9	4	0

Dendrogram

A B C D

Average Linkage Clustering

Average linkage distance:

$$D(C_i, C_j) = \frac{1}{|C_i| \cdot |C_j|} \sum_{x \in C_i} \sum_{y \in C_j} d(x, y)$$

Distance Matrix

	A	B	C	D
A	0	2	6	10
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Dendrogram



Average Linkage Clustering

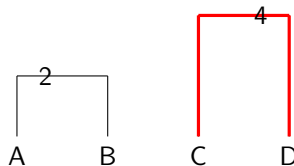
Average linkage distance:

$$D(C_i, C_j) = \frac{1}{|C_i| \cdot |C_j|} \sum_{x \in C_i} \sum_{y \in C_j} d(x, y)$$

Distance Matrix

	AB	C	D
AB	0	$\frac{6+5}{2} = 5.5$	$\frac{10+9}{2} = 9.5$
C	5.5	0	4
D	9.5	4	0

Dendrogram



Average Linkage Clustering

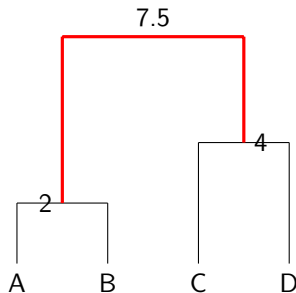
Average linkage distance:

$$D(C_i, C_j) = \frac{1}{|C_i| \cdot |C_j|} \sum_{x \in C_i} \sum_{y \in C_j} d(x, y)$$

Distance Matrix

	AB	CD
AB	0	$\frac{6+10+5+9}{4} = 7.5$
CD	7.5	0

Dendrogram



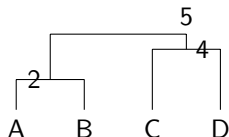
Comparison of Linkage Methods

Distance Matrix

	A	B	C	D
A	0	2	6	10
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C	6	5	0	4
D	10	9	4	0

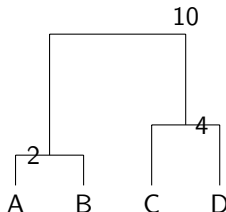
Single

$$D_{\min} = \min d(x, y)$$



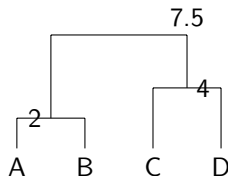
Complete

$$D_{\max} = \max d(x, y)$$



Average

$$D_{\text{avg}} = \frac{1}{n_i n_j} \sum d$$



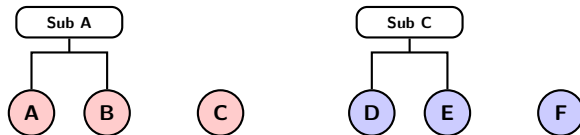
Final merge heights: single 5, complete 10, average 7.5.

The Dendrogram: A Visual Hierarchy



Leaves: Individual data points.

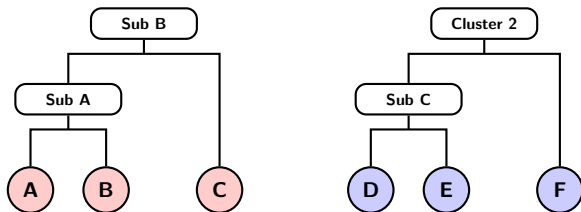
The Dendrogram: A Visual Hierarchy



Leaves: Individual data points.

Branches: Groups formed as we merge similar data.

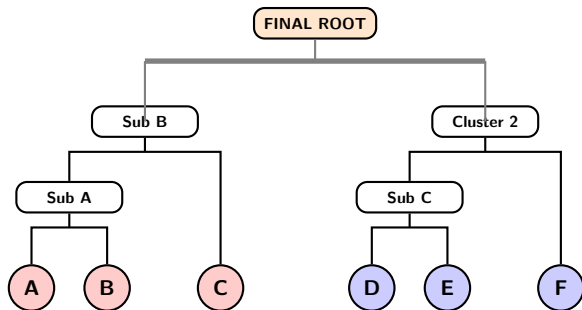
The Dendrogram: A Visual Hierarchy



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The Dendrogram: A Visual Hierarchy

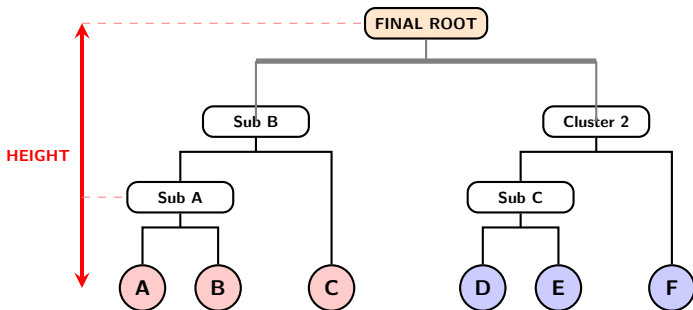


Leaves: Individual data points.

Branches: Groups formed as we merge similar data.

Root: The final connection where everything is one.

The Dendrogram: A Visual Hierarchy



Leaves: Individual data points.

Branches: Groups formed as we merge similar data.

Root: The final connection where everything is one.

Height: The distance at which clusters merge.

Conclusion

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- Hierarchical clustering builds a **tree of clusters**.
- Two main types: **Agglomerative** (bottom-up) and **Divisive** (top-down).
- **Linkage criteria** define how we measure cluster distance.
- The **dendrogram** is a powerful visualization tool.
- Widely used in bioinformatics (e.g., phylogenetics), finance (e.g., portfolio analysis), and marketing (e.g., customer segmentation).

